- Yes, the usual notion of quantum state is incomplete. There are "hidden variables" that, if only we could access them, would allow complete predictability. There are two versions of this view. In version *A*, the hidden variables are hard to measure but in principle they are experimentally available to us. In version *B*, because we are made of quantum mechanical matter and therefore subject to the restrictions of quantum mechanics, the hidden variables are, in principle, not detectable.

- No, the hidden variables concept does not lead us in a profitable direction. Quantum mechanics is unavoidably unpredictable. Quantum mechanics is as complete a calculus of probabilities as is possible. The job of a physicist is to learn and use this calculus.

I don't know what the ultimate answer to this question will be, or even if it will prove to be a useful question. But for our purposes, it's not important what any particular physicist believes about the ultimate meaning of the quantum state. For practical reasons, we will adopt the second view.

In practice, what this means for the quantum spin of Lecture 1 is that, when the apparatus $\mathcal{A}$ acts and tells us that $\sigma_z = +1$ or $\sigma_z = -1$, there is no more to know, or that can be known. Likewise, if we rotate $\mathcal{A}$ and measure $\sigma_x = +1$ or $\sigma_x = -1$, there is no more to know. Likewise for $\sigma_y$ or any other component of the spin.

## 2.2 REPRESENTING SPIN STATES

Now it's time to try our hand at representing spin states using state-vectors. Our goal is to build a representation that captures everything we know about the behavior of spins. At this point, the process will be more intuitive than formal. We will try to fit things together the best we can, based on what we've already learned. Please read this section carefully. Believe me, it will pay off.